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Subject: Physics 1 **Paper Code: BS-PH201**

Technical Report Writing

On

Fundamentals of Harmonic Oscillation in Mechanical Systems

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🧠 **Abstract:** This report explains the fundamentals of harmonic oscillation in mechanical systems such as springs and pendulums. It includes theoretical concepts, equations of motion, and practical applications, along with solutions to numerical problems related to simple harmonic motion and oscillatory systems.

📌 **Introduction:** Harmonic oscillation is a fundamental concept in physics that describes periodic motion where the restoring force is directly proportional to displacement and acts toward the equilibrium position. This type of motion, known as Simple Harmonic Motion (SHM), is widely observed in mechanical systems such as spring-mass systems, pendulums, and vibrating structures.

The study of harmonic oscillation is essential in engineering and physics because it helps analyze systems involving vibrations, waves, and stability. Understanding SHM allows engineers to design systems that can withstand oscillations or use them effectively, such as in suspension systems, clocks, and electrical circuits.

This report aims to explore the principles of harmonic motion, derive important equations, and apply them to real-world problems. It also discusses damped and forced oscillations, resonance phenomena, and oscillatory circuits. The report is structured into theoretical explanations, problem-solving sections, and conclusions.

💡 **Procedure and Discussion:**

🔷 **Simple Harmonic Motion (SHM)**

SHM is defined as motion in which acceleration is proportional to displacement and directed towards the mean position: $a = -\omega^2 x$

🔷 **Question 1: A body executing SHM has velocities 80 cm/sec. and 60 cm/sec when displacements are 3 cm and 4 cm respectively. Calculate the amplitude and frequency of vibration.**

Given: $v_1 = 80$ cm/s at $x_1 = 3$ cm

$v_2 = 60$ cm/s at $x_2 = 4$ cm

Using: $v^2 = \omega^2 (A^2 - x^2)$

Solving: $80^2 = \omega^2 (A^2 - 9)$

$60^2 = \omega^2 (A^2 - 16)$

Subtracting: $2800 = 7\omega^2$

$\Rightarrow \omega^2 = 400$

So, $\omega = 20$ rad/s

Substitute: $6400=400(A^2-9)$

$\Rightarrow A^2 = 25$

$\Rightarrow A = 5$ cm

Frequency: $f = \frac{\omega}{2\pi} = \frac{20}{2\pi} \approx 3.18$ Hz

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🔷 Free and Forced Vibrations

🔵 Free vibration:

$$m\frac{d^2x}{dt^2} + kx = 0$$

🔵 Forced vibration:

$$m\frac{d^2x}{dt^2} + kx = F_0 \sin(\omega t)$$

Where:

- m = mass
- k = spring constant
- F_o = external force

🔷 Floating Cube Oscillation

When displaced, buoyant force acts as restoring force:

$$F = -kx$$

Thus, motion is SHM.

Frequency:

$$F = \frac{1}{2\pi}\sqrt{\frac{\rho g}{dL}}$$

🔷 Force Constant & Energy in SHM

Force constant (k): stiffness of system

$$F = -kx$$

Total energy: $E = \frac{1}{2}kA^2$

This remains constant → proves energy conservation.

🔷 RLC Circuit

Given: R = 100 Ω, L = 10 mH, C = 100 µF

Natural frequency:

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(10\times10^{-3})(100\times10^{-6})}} = 1000 \text{ rad/s}$$

Since R is small → system is **oscillatory (underdamped)**.

🔷 Resonance

- **Amplitude Resonance:** Maximum amplitude at resonance frequency
- **Velocity Resonance:** Maximum velocity

🔷 Logarithmic Decrement

It measures damping:

$$\delta = \ln\left(\frac{x_1}{x_2}\right)$$

🔷 Damped Oscillation Equation

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0$$

For underdamped motion:

$$X = Ae^{-\gamma t}\cos(\omega t + \varphi)$$

🧩 **Conclusion:** This report highlights the principles of harmonic oscillation and their applications in mechanical and electrical systems. It demonstrates how SHM governs various natural and engineered systems. Understanding oscillatory behavior is crucial for analyzing vibrations, resonance, and system stability in engineering.

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